

Swapnil Dubey

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EDUCATION

The University of British Columbia

BSc in Computer Science and Statistics (Dean's Honour List), Science Co-op, GPA: 3.7/4.0

Sep 2022 – May 2027

Vancouver, BC

WORK EXPERIENCE

Software Engineer Co-op

British Columbia Investment Management Corporation (BCI)

Jan 2026 – Apr 2026

Vancouver, BC

- Owned end-to-end delivery of an internal platform serving **70+ Legal Affairs users** — gathered requirements, defined acceptance criteria, and managed **dev/staging/production** releases across the full SDLC using **Python, SQL, PowerApps**
- Saved **~3 hrs/week** for 100+ stakeholders by scoping and shipping a **Python LLM/RAG pipeline** on Azure — translated a manual summarization workflow into an automated, stakeholder-approved solution
- Eliminated manual data entry across 2 legal trackers by building an email parsing pipeline that extracted structured JSON via **Claude API** and wrote classified records directly to SharePoint
- Unified 3 siloed systems into a single access layer for **100+ Legal Affairs users** by integrating external REST APIs and Azure SQL — cutting ad-hoc data requests to the team and reducing sync prep time **~60%**

Business and Cyber Risk Analyst Co-op

Teck Resources Ltd.

May 2025 – Dec 2025

Vancouver, BC

- Served as primary PM/POC for a **ServiceNow intake redesign** for 10K+ employees — defined services, request types, SLAs, and evidence requirements; managed backlog and documentation in **Jira and Confluence**
- Lifted phishing report rate **~20%** across 10K+ employees by designing A/B tests, applying send-time and segmentation analysis, and presenting findings at **quarterly board readouts** for VP, CFO, and CISO
- Cut report prep **~4 hrs/month** for VP and Board governance cycles by joining LMS, phishing, and finance datasets via **SQL and Power Query** to produce KPI roll-ups and MoM variance analyses
- Eliminated **10+ manual entries/week** and improved enterprise audit traceability by shipping a **Power Automate** workflow that mirrored ServiceNow intake fields directly to SharePoint risk tracking lists

TECHNICAL PROJECTS

Fraud Detection MLOps Pipeline | Python, XGBoost, MLflow, FastAPI, Docker, GitHub Actions, Evidently

Mar 2026

- Framed the problem, defined success metrics (AUPRC, F2 threshold, drift triggers), and evaluated model tradeoffs (XGBoost vs. LR on imbalanced data) — achieved **0.874 AUPRC** (21% above baseline) on **590K+ transactions**
- Decoupled model updates from API code via **MLflow** alias-based champion promotion; containerized inference with **Docker**, automated CI/CD via **GitHub Actions**, and monitored production drift with **Evidently**

Music Streaming Revenue Forecasting | Python, pandas, Prophet, ARIMA, statsmodels, matplotlib

Feb 2026

- Scoped and built a revenue accrual forecasting engine for Finance — benchmarked ARIMA, Prophet, and Exponential Smoothing to justify an RPS-based approach given 8-month data constraints across **5 DSPs, 10 territories, 4 BUs**
- Eliminated forecast bias by detecting structural regime changes (YouTube RPS shift, TikTok monetization decline) via CoV analysis; delivered **Finance-ready output** with conservative/aggressive scenarios and 3 strategic insights

Prescripto – AI Healthcare Transcription | Python, Whisper API, Gemini 2.0 Flash, Claude API

Mar 2025

- Defined the product problem (language barriers + admin overhead in clinical settings), scoped AI-assisted solution, and shipped a speech-to-summary pipeline supporting **50+ languages** — reducing SOAP note time **35%** (143 ms/request)
- Cut critical medical hallucinations **40%** by implementing a dual validation layer combining regex and **Claude-3** semantic checks against ICD-11 codes

LEADERSHIP

Section Leader

Stanford University Code in Place

Apr 2024 – May 2024

Remote

- Achieved **90% retention rate** mentoring **20+ students** in Python; boosted cohort engagement **15%** through collaborative curriculum refinements with fellow instructors

SKILLS

Languages & Databases: Python; R; SQL (Azure SQL, PostgreSQL, Oracle, MySQL); JavaScript; HTML/CSS; REST APIs; Git

Product & Analytics: Requirements gathering; User stories; A/B testing; KPI design; Stakeholder readouts; Jira; Confluence; Agile/Scrum

AI & ML: Scikit-learn; XGBoost; MLflow; Evidently; LLMs (Claude, OpenAI, Cohere); RAG pipelines; Prompt engineering

Tools & Platforms: Docker; GitHub Actions; FastAPI; Power Automate; Power BI; Power Apps; SharePoint; Microsoft Azure; GCP